Adrenal & Retroperitoneum

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 145	Surgical & Radiographic Anatomy of the Adrenals
Ch 146	Adrenal Disorders: Pathophysiology, Evaluation & Medical Management
Ch 147	Surgery of the Adrenal Glands
Ch 140	Retroperitoneal Tumours
★	Putting it together: adrenal and retroperitoneal decisions

145 Surgical and Radiographic Anatomy of the Adrenals

- P1 Describe anatomy of arenal glands AND steps in production of adrenal hormones & perioperative management of a patient of pheochromocytoma — *relations, arteries, veins, zones, embryology*

Not in the paper answer:

- Neonatal weight, fetal zone, rests (§1); normal imaging (§5).

1 · Embryology

The shared origin with the gonad explains adrenal rests.

- On the adrenogonadal ridge, the cephalic part forms cortex and the caudal part gonad. Neural crest enters at weeks 8–10.
- At birth the gland is **2×** adult weight. The **fetal zone involutes in year 1**, replaced by glomerulosa (ZG) and fasciculata (ZF) under adrenocorticotropic hormone (ACTH). Zona reticularis (ZR) reaches adult volume at 6–8 y.
- In **renal agenesis** the adrenal is **discoid**.
- **Adrenal rests** occur in ≈1% of adults (near coeliac axis) and along gonadal descent in 7.5–15% of neonates (regress). In congenital adrenal hyperplasia (CAH) they form **testicular masses**.

2 · Zones and hormones

Each zone makes one hormone class.

Cortex ≈ 90%, medulla ≈ 10%. †Outside inwards: **GFR = salt, sugar, sex**.

- Glomerulosa ~15% (aldosterone; small clustered cells); fasciculata **~80%** (cortisol; lipid-laden "clear cells"); reticularis 5–7% (DHEAS, dehydroepiandrosterone sulphate; anastomosing cords); medulla ~10% (chromaffin cells).
- Medulla is a **modified sympathetic ganglion**; chromaffin cells are postganglionic neurones without axons.

3 · Relations and blood supply

These set exposure and venous risk.

- ≈5 g each, equal in both sexes. Right is **retrocaval**, directly above the upper pole; the left lies medial to the upper pole.
- **Exposure**: right, liver up, inferior vena cava (IVC) medial, duodenum and kidney down. Left, pancreas and spleen up, splenic flexure medial.
- Three arteries give 10–50 capsular twigs: **many in, one vein out**. Cortical blood drains into the medulla, so cortisol drives **epinephrine synthesis**.
- The **right vein** is short and enters the **posterior IVC**, so it can avulse. The **left** enters the left renal vein, joined by the inferior phrenic.

> **KEY POINT**
> Right side is riskier: high, retrocaval, short vein with little working length; avulsion tears the cava.

5 · Imaging appearance

Normal appearances, to spot abnormal.

- **CT** (computed tomography) has the best spatial resolution: an inverted **V/Y**, limbs 3–6 mm (thinner than crus). Unenhanced **≤10 HU** means lipid-rich. It shows morphology only.
- **MRI** has the best contrast; coronal views confirm origin. T1 is slightly below liver; on **T2 it resembles fat**.
- **Ultrasound** is hypoechoic, for solid versus cystic.
- **Angiography**: adrenal venous sampling (AVS) is the only **functional** study; cavography shows intraluminal tumour; arteriography rarely.

146 Pathophysiology, Evaluation and Medical Management of Adrenal Disorders

1 · Applied anatomy and physiology

A few physiological facts explain most test results.

- P1 Describe anatomy of arenal glands AND steps in production of adrenal hormones & perioperative management of a patient of pheochromocytoma — *veins, arteries, zones, PNMT, COMT forms metanephrines*

Not in the paper answer:

- Anomalous veins (9%); aldosterone spared in pituitary failure; volume, not sodium, rises.
- Anomalous venous drainage occurs in up to **9%**, which matters for adrenal vein sampling (AVS). Triple supply means a partial-adrenalectomy remnant rarely infarcts.
- **Pituitary failure or exogenous steroids spare aldosterone**, because the glomerulosa obeys angiotensin II and potassium (K^+). The inner zones atrophy without adrenocorticotropic hormone (ACTH).
- Aldosterone raises volume, **not serum sodium**.
- Phenylethanolamine-N-methyltransferase (PNMT) is almost unique to the medulla, so mainly adrenal tumours make epinephrine. >90% of circulating metanephrine is adrenal.
- α2 suppresses insulin: hyperglycaemia, then rebound hypoglycaemia after resection.

2 · Evaluation of an adrenal mass – principles

Every adrenal mass faces two questions.

> **KEY POINT**
> Two questions: **(1) Is it malignant?** (imaging). **(2) Is it hormonally active?** (biochemistry). Yes to either means treat (usually resect). Imaging cannot identify functional adenomas, so biochemistry is never omitted.

Category	Indications for adrenalectomy (Box 146.1)
Functional	Autonomous cortisol secretion; phaeochromocytoma; lateralised aldosterone excess
Oncological	>4 cm (except myelolipoma); suspicious imaging (lipid-poor, heterogeneous, irregular, infiltrating); growth >1 cm on follow-up; isolated metastasis (multidisciplinary)
Myelolipoma	Very large and/or symptomatic
At nephrectomy for renal cell carcinoma (RCC)	Adrenal abnormal or not visualised; thrombus to adrenal vein level
Bilateral	Failed pituitary surgery for Cushing disease; selected ectopic ACTH; AIMAH; PPNAD (§7)

3 · Imaging

Imaging looks for lipid, then for washout.

- P2 P4 Algorithm for management of an incidentaloma; Adrenal mass evaluation and adrenocortical carcinoma: diagnosis and management — *APW/RPW formulas, size–malignancy risk, ESE–ENSAT follow-up*

Not in the paper answer:

- ≤0 HU 100% specific; 7% of metastases <10 HU; RCC/hepatoma metastases wash out.
- Surgical-series rates; CT underestimates size; calcification in 30% of ACC.

- **Unenhanced computed tomography (CT)**: **≤10 Hounsfield units (HU)** means a lipid-rich adenoma (no known malignancy), >70% of adenomas. ≤0 HU is 100% specific but insensitive.
- Up to **7%** of metastases are <10 HU, so every lesion still needs biochemistry.
- **Chemical-shift magnetic resonance imaging (MRI)**: signal drop-out is about as accurate as unenhanced CT.
- **CT washout** is the gold standard for lipid-poor lesions (≈30% of adenomas), with phases at 0, 60 s and 15 min. It is an adenoma if the absolute percentage washout (APW) is **>60%** or the relative washout (RPW) **>40%**. RCC and hepatoma metastases can also wash out.

> **ALGORITHM**
>
> 1. Unenhanced CT: if ≤10 HU or MRI drop-out, then adenoma.
> 2. If >10 HU, then do a 15-min washout CT.
> 3. If APW >60% / RPW >40%, then it is an adenoma; if below, an adenoma is very unlikely, so manage as indeterminate.

Size	Malignancy (large unselected series)	Surgical series
<4 cm	0.1%	≈2–5% adrenocortical carcinoma (ACC)
4–6 cm	2.4%	≈6–10%
>6 cm	19.5%	≈25%

- **4 cm** is the consensus threshold (sensitive, poorly specific); 6 cm is more specific. CT underestimates pathological size.
- **Benign imaging overrides size**: no <10 HU lesion was malignant in the largest series. Age modifies the decision.
- Suspicious: heterogeneity, irregular margins, an enhancing rim and invasion. **Macroscopic fat** (<−30 HU) means myelolipoma. **Calcification is not reassuring**, as it occurs in up to 30% of ACC.
- Classic follow-up re-images at 6, 12 (with or without 24) months; growth of ≥1 cm means consider resection.

Functional imaging

- P2 MIBG scan — *isotopes; best in MEN-2; cold in SDHB/VHL*

Not in the paper answer:

- FDG PET poor for RCC/hepatoma metastases; contrast is safe.

- **Fluorodeoxyglucose positron emission tomography (FDG PET)** detects malignancy (ACC, metastasis) when CT is inconclusive. It stages phaeochromocytoma and paraganglioma (PPGL), best for metastases, succinate dehydrogenase B (SDHB) and von Hippel–Lindau (VHL); poor in multiple endocrine neoplasia type 2 (MEN-2) and for RCC or hepatoma metastases.
- For phaeochromocytoma >5 cm, do metaiodobenzylguanidine (MIBG) or FDG PET pre-op to exclude metastases. Iodinated contrast does **not** precipitate crisis.

4 · Hormonal work-up (all lesions >1 cm)

Imaging cannot show function, so test every lesion.

> **KEY POINT**
> Every incidentaloma: **1-mg dexamethasone suppression test (DST) and metanephrines**. Add an **aldosterone-to-renin ratio (ARR)** if there is hypertension or unexplained hypokalaemia. >10% are functional. **Never biopsy, sample or excise before excluding phaeochromocytoma**. Canadian Urological Association (CUA) guidance allows omitting metanephrines in asymptomatic <10 HU lesions.

Incidentalomas comprise cortisol adenoma 5%, phaeochromocytoma 5%, aldosteronoma 1%, ACC 5% and metastasis 2.5%, so **≈18%** are potentially surgical.

Cortisol

- P2 Cushing syndrome and the dexamethasone suppression test — *DST cut-offs and pitfalls, MACS*

Not in the paper answer:

- 5.0 cut-off misses 18%; UFC unreliable if GFR <60.
- 1-mg DST: **≤1.8 µg/dL (50 nmol/L)** excludes autonomy; 1.8–5.0 means possible autonomy; **>5.0 µg/dL (140 nmol/L)** is >95% specific. The 5.0 cut-off misses up to 18% of Cushing syndrome.
- Urinary free cortisol (UFC) is insensitive for mild excess and unreliable if the glomerular filtration rate (GFR) is <60 mL/min.

Catecholamines

	Plasma free metanephrines	24-h urinary fractionated metanephrines
Performance	Sensitivity 97–99% – best rule-out	>97% by tandem mass spectrometry
Conditions	**Supine after 20 min rest**, fasting	Normal renal function
Highly indicative	Normetanephrine >2.2 nmol/L and/or metanephrine >1.2 nmol/L (≈4× upper limit of normal)	—

- **False positives**: paracetamol, mesalazine, sulfasalazine (stop ≥5 days), tricyclics, **phenoxybenzamine**, severe illness and seated sampling.
- If equivocal, repeat supine off β-blockers, check chromogranin A (less specific if GFR <80 mL/min), or do a **clonidine suppression** test.

5 · Biopsy and incidentaloma management

Biopsy is rarely useful; most lesions need nothing.

- Biopsy answers only "is this a metastasis?" (cytology cannot separate adenoma from carcinoma). The European Society of Endocrinology–European Network for the Study of Adrenal Tumors (ESE–ENSAT) advises **no biopsy without a history of malignancy**. Complications 3–12%; seeding in ACC.

ALGORITHM

1. If imaging characterises the lesion, then do not biopsy.
2. If it is functional, then do not biopsy.
3. If ACC is suspected, then do not biopsy.
4. If the result will change management (e.g., known primary) and justify risk, then biopsy, **after excluding phaeochromocytoma**.

- Incidentalomas appear on ≈5% of scans. Adenomas are >85%; ≈93% are inactive and need no treatment.
- Repeat hormone testing: none (ESE) versus annual (CUA and the American Association of Clinical Endocrinology, AACE). New hormone excess develops in <1–1.7%.

6 · Primary aldosteronism (PA)

Commonest curable hypertension; lateralisation decides surgery.

- P2 How will you evaluate and manage 38 yr male with uncontrolled hypertension — *ARR, saline, CT, AVS; cure ≈94%*

Not in the paper answer:

- Normokalaemia 63–91%, escape, stroke/MI/AF excess; FH-I and FH-III.
- Screening list, drug effects on ARR, other confirmatory tests, when to skip confirmation.
- AVS selectivity and lateralisation ratios; post-op hyperkalaemia; SPARTACUS.

- PA is the commonest secondary hypertension (3–13% of hypertensives). Renin is suppressed (high in secondary aldosteronism).
- There is no hypernatraemia or oedema because of **mineralocorticoid escape**. Most patients (63–91%) are **normokalaemic**.
- Versus essential hypertension: stroke ≈4×, myocardial infarction (MI) ≈6.5× and atrial fibrillation (AF) ≈12×.

Subtype	%	Surgically curable?
Bilateral (idiopathic) hyperplasia	60	No – mineralocorticoid receptor antagonist (MRA)
Aldosterone-producing adenoma	35	Yes
Unilateral hyperplasia	2	Yes
Carcinoma, ectopic (ovarian), familial hyperaldosteronism (FH)	<1 each	Carcinoma/ectopic yes; FH no

- **FH-I** (glucocorticoid-remediable) is autosomal dominant (AD). A chimeric gene joins the CYP11B1 promoter to CYP11B2 coding sequence, so aldosterone comes under ACTH control. Early severe hypertension, young stroke; treat with glucocorticoid with or without MRA.
- **FH-III** (KCNJ5): very young, bilateral hyperplasia, severe hypokalaemia.
- **Screen (Endocrine Society 2016)**: resistant hypertension (≥3 drugs including a diuretic) or control on ≥4 drugs; BP ≥150/100 on three days; hypertension and hypokalaemia; **incidentaloma with hypertension**; sleep apnoea; onset <20 y or stroke <50 y; family history of PA; disproportionate target-organ damage.
- **ARR**: morning, seated, plasma aldosterone concentration (PAC, ng/dL) divided by plasma renin activity (PRA, ng/mL/h). Positive if **ARR >20–30 with PAC >15–20 ng/dL**. Floor PRA at 0.2; correct K^+.

Drug	Effect on ARR	Action
Spironolactone, eplerenone, amiloride	Profound	Stop ≥4–6 weeks; use α1-blocker or calcium channel blocker (CCB)
β-blockers	Lower renin, so **false positive**	May continue (or stop)
Angiotensin-converting enzyme (ACE) inhibitors / angiotensin receptor blockers (ARBs)	Raise renin, so false negative	Continue; undetectable renin is suspicious
Calcium channel blockers	Neutral	Continue

Confirmatory test	Protocol	Positive
Oral sodium loading	3 days high salt; 24-h urine Na >200 mmol	Urine aldosterone >12 µg/day
IV saline infusion	2 L 0.9% saline over 4 h, recumbent	PAC >5 ng/dL (>10 suggests adenoma)
Captopril challenge	25–50 mg	PAC stays >15 ng/dL – when salt load contraindicated

- **Skip confirmation** if there is spontaneous hypokalaemia, undetectable renin and PAC >20 ng/dL. Only 50–87% of positive screens confirm.
- CT all surgical candidates (excludes ACC), but ≈20% of aldosteronomas are <1 cm, so CT alone misdirects ≈25%.
- **AVS** (sensitivity 95%, specificity 100% if done well): K^+ corrected, morning, both adrenal veins and inferior vena cava (IVC), measuring aldosterone and cortisol.
- **Selectivity** proves the catheter is in the vein: adrenal vein to IVC cortisol >1.1–5:1 (higher with cosyntropin). **Lateralisation**: a cortisol-corrected aldosterone-to-cortisol (A/C) ratio **>2–4:1** indicates ipsilateral adrenalectomy.
- **Omit AVS** if age <40 with a clear unilateral adenoma >1 cm and a normal contralateral gland, or if ACC is suspected. Genetic testing if onset <20 y or family history. If AVS fails: NP59, posture test.

ALGORITHM

1. If there is a screening indication, measure the ARR.
2. If the ARR is positive, confirm with a suppression test. If there is a family history or onset <20 y, test for FH-I.
3. If not a surgical candidate, then MRA.
4. If a surgical candidate, do CT. If age <40 with a unilateral >1 cm nodule, then adrenalectomy.
5. Otherwise do AVS: if lateralised, laparoscopic adrenalectomy; if not lateralised, MRA.

- Residual hypertension: older, male, obese, long duration.
- After surgery, check PAC and ARR; † stop K^+ and MRA. High-salt diet; watch for **hyperkalaemia** for weeks (the contralateral glomerulosa is suppressed). BP settles over 6 months.
- MRA: spironolactone 25–50 mg, up to a maximum of 400 mg (gynaecomastia, erectile dysfunction); eplerenone 25 to 100 mg (selective).

7 · Cushing syndrome and cortisol excess

Urologists meet it as stones, a mass or bilateral adrenalectomy.

- P2 Cushing syndrome and the dexamethasone suppression test — *causes, IPSS, Nelson, pituitary cure*

Not in the paper answer:

- Ectopic CRH; occult source in 1/3; macroadenoma cure <15%.
- AIMAH and PPNAD; insufficiency ≈65% after MACS; HPA recovery times.
- Ectopic ACTH also arises from thymic, islet cell and medullary thyroid tumours; occult in up to 1/3. Ectopic corticotropin-releasing hormone (CRH) causes <1%.
- **Urological**: hypogonadotropic hypogonadism and **urolithiasis in up to 50%**, so test cushingoid stone formers.

ALGORITHM

1. Exclude exogenous steroid.
2. Screen with 1-mg DST, late-night salivary cortisol (LNSC) ×2 or UFC ×2.
3. If abnormal, endocrinologist confirms; measure **morning ACTH**.
4. If ACTH low: adrenal, do CT.
5. If normal/high: pituitary MRI and **inferior petrosal sinus sampling after CRH**. A central-to-peripheral gradient means Cushing disease; no gradient means ectopic.

- Cushing disease: macroadenoma cure <15%, relapse up to 25%; radiotherapy is slow.
- **Bilateral adrenalectomy** is for failed pituitary treatment, life-threatening hypercortisolism, or unresectable or occult ectopic ACTH (the primary is resectable in only ≈10%). Cost: lifelong replacement, crisis risk and **Nelson syndrome** (chiasmal compression, oculomotor palsies).

	ACTH-independent macronodular adrenal hyperplasia (AIMAH)	Primary pigmented nodular adrenocortical disease (PPNAD)
Glands	Massive, nodules up to 4 cm, >60 g each	Normal size; small black/brown nodules in atrophic cortex
Association	McCune–Albright	Carney complex (≈half): spotty pigmentation, Sertoli cell tumours
Clue	Looks ACTH-driven – check ACTH	Paradoxical rise in UFC after dexamethasone
Treatment	Bilateral adrenalectomy; unilateral removal of larger gland has worked in AIMAH	

- **Mild autonomous cortisol secretion (MACS)**: individualise surgery for hypertension, obesity, dysglycaemia or osteopenia. Progression to overt Cushing is very unlikely.
- **Post-op adrenal insufficiency is expected**, because the contralateral gland is atrophic: ≈65% after MACS and ≈99.7% after overt Cushing.
- Hydrocortisone 100 mg IV on the day of surgery, then oral taper (25–50 mg/day). Alternative: dexamethasone 0.5 mg 6-hourly; morning cortisol <50 nmol/L confirms cure, then switch to hydrocortisone. Maintenance is 10–12 mg/m²/day in 2–3 doses.
- The hypothalamic–pituitary–adrenal (HPA) axis recovers in a median of 6.5 (MACS) to 11 (Cushing) months.

8 · Phaeochromocytoma and paraganglioma (PPGL)

These can kill on the table, so block before surgery.

- P2 Phaeochromocytoma: overview, pre-operative and intra-operative management — *genetics, blockade doses, targets, post-op care, lifelong metanephrines*

Not in the paper answer:

- 24% of "sporadic" cases germline; paediatric rates; gene loci, VHL subtypes, SDHD imprinting.
- Standing systolic target; metyrosine dose; malignancy risk factors and ≈50% 5-year survival.
- 75–90% are adrenal; paraganglioma is extra-adrenal (organ of Zuckerkandl, bladder, para-aortic; head and neck ones are usually non-functional). PPGL causes ≈0.5% of hypertension.
- Rule of 10s today: ≈24% of apparently sporadic cases carry germline mutations. Malignancy ≤5% sporadic adrenal, >1/3 extra-adrenal; children up to 40% familial, >20% bilateral.

KEY POINT
The rule of 10s is outdated – every patient with PPGL should be offered genetic counselling/testing (Endocrine Society 2014).

- Headache occurs in 60–90%; paroxysmal hypertension in 30–50%.

Syndrome	Gene	Features	PPGL risk / malignancy
MEN-2A	RET 10q11.2	Medullary thyroid cancer (MTC), hyperparathyroidism	50% / ≈3%; adrenal, bilateral
MEN-2B	RET	MTC, mucosal neuromas, marfanoid	50% / ≈3%
VHL type 2	VHL 3p25	RCC, haemangioblastoma, epididymal cystadenoma	10–20% / ≈5%; normetanephrine only
Neurofibromatosis 1 (NF1)	NF1 17q11	Neurofibromas, café-au-lait	1% / ≈11%
PGL-4	**SDHB** 1p36	Abdominal/thoracic paragangliomas	≈20% / **30–50%**
PGL-1	SDHD 11q23 (**maternal imprinting** – paternally transmitted)	Head and neck, multifocal	≈20% / <3%

All are autosomal dominant. VHL subtypes: 1 has no phaeochromocytoma; 2A has phaeo but no RCC; 2B has phaeo and RCC; 2C has phaeo only.

- Measure metanephrines even in a suspected solitary metastasis. On CT the mean is ≈35 HU; the T2 "light bulb" sign is unreliable.
- **Block all with abnormal biochemistry, even if normotensive.** Phenoxybenzamine causes hypotension after resection; α1-selective drugs (including terazosin) do so less. CCBs can be sole therapy in mild cases, without reflex tachycardia.
- **Metyrosine** 250 mg 2–3 times a day, increased to 1.5–2 g, is added to α-blockade for large or metastatic tumours; it causes sedation, depression, galactorrhoea and **extrapyramidal** effects.

KEY POINT
Why never β before α? β-blockade removes β2-mediated arteriolar dilatation. Catecholamines then act unopposed on α1 receptors and cause a hypertensive crisis. Hence use selective β1 agents.

- **Targets**: seated BP 120–130/80; standing systolic >80–90 mmHg; † heart rate 60–70/min. **Volume repletion** is key.
- In **MEN-2 and VHL**, do **cortical-sparing partial adrenalectomy**; recurrence is possible and replacement may still be needed.

- Malignancy is defined **only by metastases** (bone, lung, liver, nodes). Risk: SDHB, extra-adrenal, >5 cm, dopamine-secreting. Ablation and embolisation also used; cyclophosphamide–vincristine–dacarbazine (CVD) responses are short. 5-year survival is ≈50%.
- Check metanephrines at ≈6 months, then annually for life, because recurrence occurs even >15 years later. Persistent elevation calls for MIBG.

9 · Non-functional benign lesions

Benign, but still test hormones.

Lesion	Key features	Management
Myelolipoma	Fat and haematopoietic tissue; macroscopic fat diagnostic; calcification >25%; may exceed 10 cm	Exception to 4-cm rule; still do hormonal work-up. Excise if symptomatic or large (bleeders 10–14 cm; 6–10 cm cut-offs unvalidated); embolise a bleed. Do not confuse with liposarcoma (infiltrative)
Cysts	Endothelial 45%, pseudocyst 39% (harbours malignancy ≈7%), epithelial 9%, hydatid 7%; >30 HU means haemorrhage	Endocrine work-up. **Excise if >5 cm, thick-walled, heterogeneous or symptomatic**; no decortication
Ganglioneuroma	Young; encases vessels without narrowing them	Benign; diagnosed at resection
Oncocytoma	Rare; large; ≈10% functional; no central scar	Resect; >30% malignant (unlike renal)

10 · Adrenocortical carcinoma (ACC)

Complete first resection is the only cure.

- P4 Adrenal mass evaluation and adrenocortical carcinoma: diagnosis and management — *syndromes, CT, Weiss, ENSAT, R0 surgery, mitotane, trials, surveillance*

Not in the paper answer:

- Stage distribution at diagnosis; stage IV survival <5%; 60–80% recurrence; paediatric ACC.
- Cortisol secretion signals a poor prognosis.
- Mitotic rate grades ACC and best predicts outcome; use alternative criteria for paediatric and oncocytic tumours.

ENSAT stage	Definition	At diagnosis
I	T1 (≤5 cm) N0 M0	3–4%
II	T2 (>5 cm) N0 M0	29–46%
III	T3 (periadrenal fat) or T4 (adjacent organ) or N1 or venous thrombus	11–19%
IV	Distant metastasis only	39–49%

- Overall 5-year survival 20–47%; stage IV <5%. Recurrence 60–80% after resection.
- Resect recurrences if low grade with a long disease-free interval.
- † Mitotane often needs higher-than-usual glucocorticoid doses, because it raises binding globulins and speeds clearance.
- **Paediatric ACC** is >90% functional (virilisation); test for TP53; prognosis is better. Neuroblastoma: see *Ch 63*.

11 · Adrenal metastases

In cancer patients, usually a metastasis.

- With known malignancy, **>50%** of new adrenal lesions are metastases. RCC and hepatoma deposits may wash out, so growth and biopsy decide. **Still exclude phaeochromocytoma**.
- Resect an isolated metastasis in selected patients: non-small cell lung cancer (level 1 evidence) and melanoma; RCC as part of cytoreduction.

12 · Adrenal insufficiency

Missed insufficiency after surgery can kill.

	Primary (Addison)	Secondary (pituitary/hypothalamic)
Causes	Autoimmune (anti-21-hydroxylase), **tuberculosis**, HIV-related infection, bilateral haemorrhage, infiltration, **bilateral adrenalectomy**, drugs (ketoconazole, abiraterone)	**Exogenous glucocorticoid suppression** (commonest for surgeons), pituitary tumour/apoplexy, Sheehan, radiation, critical illness
Aldosterone	Lost	Preserved
Skin	Hyperpigmentation (high ACTH and pro-opiomelanocortin)	Pallor
Electrolytes	Low Na^+, **high K^+**, high creatinine; salt craving	Low Na^+ (inappropriate ADH secretion), normal K^+
Replacement	Hydrocortisone + fludrocortisone	Hydrocortisone only

- A **crisis** mimics an acute abdomen. At risk: cortisol-secreting tumour, bilateral adrenalectomy, **previous contralateral nephrectomy/adrenalectomy**, so review old notes.
- Diagnose: morning cortisol and ACTH; ACTH stimulation. Give hydrocortisone ½–⅔ on waking.

KEY POINT

† **Adrenal crisis**: do not wait for results. Draw cortisol and ACTH, give **hydrocortisone 100 mg IV bolus**, then 200 mg/24 h (infusion or 50 mg 6-hourly). Give 0.9% saline (1 L in the first hour) with dextrose, treat the precipitant, and taper over 1–3 days. Fludrocortisone is unnecessary while hydrocortisone is >50 mg/day. Teach sick-day rules and issue a steroid card.

Surgical stress (chronic steroids)	Example	Cover
Minor	Inguinal hernia, <1 h	Hydrocortisone 25 mg
Moderate	Joint replacement	Usual dose + hydrocortisone 50–75 mg intra-op
Major	Oesophagectomy, bypass	Usual dose + hydrocortisone 50 mg IV 8-hourly for 48–72 h

Glucocorticoid	Potency	Replacement (mg/day)	Half-life (h)
Hydrocortisone	1	20	8–12
Prednisone / prednisolone	4	5	18–36
Dexamethasone	25–50	0.5	36–54

13 · Congenital adrenal hyperplasia (CAH)

May present as a mass or testicular lump.

- CAH is autosomal recessive. **21-hydroxylase deficiency is >95%**; low cortisol raises ACTH, which causes hyperplasia and androgen excess (see *Ch 59*).
- Adenomas (not ACC) occur; non-classic CAH may present as an adult mass. Testicular rests mimic tumour, so consider them before orchidectomy.

14 · Landmark trials

Trials shaping practice.

Trial	Design	Result	Take-home
SPARTACUS	RCT: CT- vs AVS-guided PA treatment	Same antihypertensive intensity at 1 y; CT and AVS discordant in half; persistent aldosteronism 20% (CT) vs 10% (AVS)	AVS remains standard for biochemical cure
FIRM-ACT	RCT: mitotane + etoposide, doxorubicin, cisplatin (EDP) vs mitotane + streptozotocin, advanced ACC	Better response and progression-free survival; overall survival similar	M-EDP first-line
ADJUVO	RCT: adjuvant mitotane vs observation, low/intermediate-risk R0 stage I–III ACC	No difference in recurrence	No routine adjuvant mitotane in low risk

15 · MCQ one-liners

MCQ ONE-LINERS

1. Commonest ectopic ACTH source: bronchial carcinoid.

147 Surgery of the Adrenal Glands

- P2 Adrenal Sx. — *indications, contraindications, preparation, approaches, steps, complications*

Not in the paper answer:

- Imaging criteria, incidentaloma rules (§2); retroperitoneoscopic landmarks (§6); metastasis, CAH, ablation (§7); partial technique (§8); outcomes (§10).

1 · Surgical anatomy (essentials)

The operation is organised around the adrenal vein.

- The **right** gland lies behind the bare area of the liver (fused in ~10%). Its vein is **<1 cm** long and enters the **posterior inferior vena cava (IVC) at 45°**; variants drain into the right hepatic or right renal vein.
- The **left** gland is more mobile (see *Ch 145*).

2 · Indications

Two gates: any functional tumour, or malignant potential.

Category	Indication
Functional (any size)	Aldosteronoma; cortisol adenoma; Cushing disease after failed trans-sphenoidal surgery; phaeochromocytoma (PCC); virilising/feminising tumour
Non-functional	Adrenocortical carcinoma (ACC); symptomatic cyst/myelolipoma; solitary metastasis
Size	**≥4 cm**; >2 cm growing ≥1 cm in 12 months
Imaging	Non-contrast computed tomography (CT) **>10 HU** (Hounsfield units); washout <50% with delayed >35 HU; chemical-shift MRI not adenoma; FDG-PET uptake equal to or above liver

KEY POINT

Trap: bilateral hyperplasia causing aldosteronism is **not** surgical, so adrenal venous sampling (AVS) comes before adrenalectomy for aldosteronism.

Incidentaloma work-up

- ACC is found in ~2%; autonomous secretion in up to 20%.
- All get a 1-mg dexamethasone suppression test (DST), never spot cortisol, and metanephrines (>2× upper limit significant).
- If hypertensive: aldosterone-to-renin ratio (ARR) >20 with aldosterone >20 ng/dL, then salt loading, then AVS. Up to 40% are normokalaemic.
- Measure sex steroids if ≥4 cm or virilised.
- Inert, ≤4 cm, ≤10 HU, homogeneous: repeat CT at 12 months. Operate on myelolipoma (≤ −20 HU) only if **>7 cm**, symptomatic or atypical, and still exclude PCC.

3 · Contraindications

Correct first, then operate.

- Low-dose aspirin may continue. In pregnancy, necessary surgery (e.g. PCC) can be done in any trimester. A prior nephrectomy may have removed the ipsilateral adrenal: confirm before contralateral adrenalectomy.

4 · Pre-operative preparation

Each tumour has its own danger.

Pathology	Preparation	Why
PCC	α-blockade for **2 weeks** (phenoxybenzamine; doxazosin, calcium channel blocker, metyrosine alternatives); β **only after α**; volume expansion	Contracted patients collapse once vein tied. ECG, echo
Cushing	†Hydrocortisone 100 mg at induction, tapered over months; thromboprophylaxis	Contralateral gland is suppressed; steroid cover dropped early mortality to zero
Aldosteronoma	†Correct K^+; control BP	†Hypokalaemia causes arrhythmia

5 · Approaches

Open surgery is for cancer.

- **Minimally invasive surgery (MIS) is standard** since 1992 (Gagner).
- The **retroperitoneoscopic** route (lateral or **prone**) avoids bowel: use after prior abdominal surgery or peritonitis. A robot suits the small space; bilateral small tumours prone without repositioning.
- **Open**: suspected **ACC**, large tumours, or invasion of the IVC or organs. Thoracoabdominal for huge tumours or IVC extension above the liver.

6 · Operative steps

Control the vein, then free the gland.

ALGORITHM

Right:

1. Elevate the liver; Kocher the duodenum; expose the IVC.
2. Dissect up from the renal vein (inferior limit) to the short adrenal vein.
3. **Warn anaesthesia**, then clip early.
4. Dissect superiorly and laterally down to psoas (posterior limit); bag the gland intact.

ALGORITHM

Left:

1. Reflect the colon at Toldt, outside Gerota; lift spleen and pancreas up.
2. Follow the left renal (or gonadal) vein to the adrenal vein; clip after warning.
3. Protect pancreas and splenic artery; divide the ganglion nexus superomedially; bag.

- **Retroperitoneoscopic**: 1-cm incision below 12th rib tip, balloon, two ports 8 cm apart. Along psoas the **renal artery is met first**. The left vein is anterocephalad to the renal artery. The right vein enters the posterolateral IVC above the hilum; avulsion causes major bleeding.

KEY POINT

Ligate the vein early in PCC to limit the catecholamine surge; hypotension follows, so tell anaesthesia first.

7 · Special pathologies

Diagnosis changes the operation.

- **PCC intra-op**: treat a surge with nitroprusside, nitroglycerin or phentolamine, and arrhythmia with esmolol.
- **ACC** needs a high-volume surgeon (>6, ideally ≥20/year) and an **open, intact, en bloc** resection.
- **Metastasis**: lung and kidney give ≈ two-thirds. Operate when the adrenal is the only disease; renal origin and <5 cm do best.
- **Congenital adrenal hyperplasia (CAH)**: bilateral adrenalectomy avoids iatrogenic Cushing, but risks crisis and testicular rest tumours.
- **Ablation** (radiofrequency, >50–60 °C) should be avoided for PCC or ACC outside trials.

8 · Partial (cortical-sparing) adrenalectomy

Keeping cortex avoids lifelong steroids.

- **Indications**: bilateral or hereditary PCC, familial syndromes, prior contralateral adrenalectomy, children, Conn adenoma and cysts.
- Arteries enter superiorly, medially and inferiorly, so **lateral dissection is safe**. The main vein may be divided because there is collateral outflow.
- Set margins by intra-op ultrasound and check remnant viability with indocyanine green. †Retain about one-third of a gland.

9 · Complications

Bleeding, then hormonal problems.

- Bleeding: pressure, raise pneumoperitoneum, extra port, side-bite clamp the IVC, convert if uncontrolled. Residual phenoxybenzamine prolongs post-PCC hypotension (volume with or without α-agonist).
- †Check glucose hourly for 24–48 h after PCC. Treat insufficiency with IV hydrocortisone 100 mg and saline. For hyperkalaemia after aldosteronoma, stop K^+ and spironolactone, occasionally adding fludrocortisone.
- Flank bulge after open surgery: protect the subcostal neurovascular bundle.

10 · Outcomes

Follow-up is lifelong for PCC.

- PCC recurs in up to **16%** over life, so check BP and metanephrines annually.
- †Aldosteronoma: hypokalaemia is corrected in nearly all; hypertension is cured in about half.
- Cushing: insulin resistance may persist. Advanced-stage ACC has a 5-year survival **<15%**.

140 Retroperitoneal Tumours

1 · Anatomy of the retroperitoneum

Anatomy predicts which neighbours a retroperitoneal (RP) tumour will involve.

- P1 Describe in brief the anatomy of the retroperitoneum. Explain the principles of laparoscopic retroperitoneal surgery while mentioning the salient steps of retroperitoneal laparoscopic nephrectomy. — *boundaries, contents, spaces, vessels, nerves at risk*

Not in the paper answer:

- Side-specific structures at risk; psoas invasion depth predicts nerve injury.
- On the right, the concerns are the duodenum, inferior vena cava (IVC), posterior diaphragm, right liver and right colonic mesentery; lost fat planes mean involvement. On the left, they are the aorta, splenic hilum, pancreatic tail and left colonic mesentery.
- **Depth of psoas invasion predicts nerve injury**. Genitofemoral injury causes thigh paraesthesia; femoral injury weakens hip flexion and knee extension, which is worse. Book specialist help in advance.

> **KEY POINT**
> The operative problem is usually the neighbouring structure, not the tumour: RP tumours grow silently (resected sarcomas average **>15 cm**) and present with multi-organ involvement.

2 · Classification

The group decides first treatment.

Group	Entities	First treatment
Germ cell / gonadal	Seminoma, nonseminoma, sex cord stromal	Platinum chemotherapy (germ cell tumour, GCT); RP lymph node dissection, RPLND (sex cord stromal)
Mesodermal — malignant	Liposarcoma, leiomyosarcoma, undifferentiated pleomorphic sarcoma (UPS), synovial, Ewing, desmoplastic small round cell tumour (DSRCT), unclassified	Surgery; chemotherapy (Ewing, DSRCT)
Mesodermal — benign/intermediate	Lipoma, leiomyoma, solitary fibrous tumour (SFT), PEComa, desmoid, myelolipoma, GIST	Excision / observation
Neurogenic (~10%)	Schwannoma, neurofibroma, malignant peripheral nerve sheath tumour (MPNST), ganglioneuroma, neuroblastoma, paraganglioma	Surgery
Haematological (~1/3 of primary RP)	Lymphoma, plasmacytoma	Chemotherapy — biopsy only
Cystic	Lymphangioma, mucinous cystadenoma, cystic mesothelioma, cystic teratoma	Complete excision
Unknown primary	Poorly differentiated carcinoma encasing vessels	Histology-led

3 · Clinical evaluation

Tissue only if it changes treatment.

- Most are painless or incidental; mass effect is late (leg oedema, ureteric or bowel obstruction, nerve pain). The sequence is **imaging, then serology, then tissue**.
- **CT chest/abdomen/pelvis** comes first (NCCN). **Negative HU (fat)** suggests liposarcoma, angiomyolipoma or myelolipoma. CT cannot tell origin or malignancy. **MRI** shows vascular involvement and suits contrast allergy or renal impairment. **PET-CT** is selective: post-chemo seminoma residue, liposarcoma grade, lymphoma.
- In a young man check **alpha-fetoprotein (AFP) and hCG** (with or without scrotal ultrasound); in an older man, PSA. Add blood count and electrophoresis (M-spike); † metanephrines if paraganglioma is possible.
- **Biopsy** only when it **changes first treatment**: lymphoma, GCT, Ewing, or high-grade sarcoma for neoadjuvant therapy. Use **core (14–16 G coaxial) rather than fine-needle aspiration (FNA)**; † take a retroperitoneal (posterior) route, with the tract inside the future resection field.
- Avoid biopsy of purely cystic lesions, † an unblocked paraganglioma, or a young man with raised AFP/hCG. Liposarcoma cores are often non-diagnostic.

4 · Mesenchymal tumours (sarcoma)

Histology predicts behaviour: some fail locally, some spread.

- Soft-tissue sarcoma is <1% of adult cancers; **15–20%** arise in the RP.

Entity	Key features	Treatment
Well-/de-differentiated (WD/DD) liposarcoma (commonest, **~60%**)	Fat density; **MDM2/CDK4 (12q) amplification**, ring chromosomes; S100+	Radical compartmental resection
Leiomyosarcoma (**~20%**)	From vessel media (IVC); no fat; SMA, desmin, h-caldesmon+	Negative margins + adjuvant
UPS (ex-MFH)	5th–6th decade; no fat, peripheral nodular enhancement	Wide resection
Unclassified (≤20%)	Exclusion; prior RT in a quarter	Screen MDM2
Synovial	Calcification; TLE1; **t(X;18)**	Chemosensitive
SFT	**NAB2-STAT6**; Doege-Potter hypoglycaemia (IGF)	En bloc; radiosensitive
Ewing	EWSR1-FLI1 t(11;22); infiltrative	Chemotherapy-based
DSRCT	Young men; peritoneal implants; WT1-EWSR1 t(11;22)	Multimodal; poor
PEComa	HMB-45 + muscle markers; TSC2	Excision; mTOR inhibitors if unresectable
Desmoid	Nuclear β-catenin; APC/CTNNB1; never metastasises; recurs in up to half	Surgery or observation

- **Liposarcoma**: DD has a higher-grade non-lipogenic component, so **dense nodules in a fatty mass mean de-differentiation** and are the biopsy target. WD recurs locally in up to 40%

but rarely metastasises; DD spreads distantly in up to 20%, with 5-year mortality ~30%. Mean time to recurrence is ~**2 years**, so surveillance is lifelong.

- If unresectable: anthracycline (modest); trabectedin (improves progression-free, not overall, survival); pembrolizumab for high tumour mutational burden. **Pleomorphic liposarcoma** is MDM2-negative, mimics carcinoma and metastasises early to lung.
- **Leiomyosarcoma** obstructs veins, urine and bowel. The spectrum runs leiomyoma, then STUMP (uncertain malignant potential), then leiomyosarcoma. Biopsy underestimates aggressiveness, and recurrence is frequent.

> **KEY POINT**
> MDM2 amplified (WD/DD) means local disease, won or lost at the margin; MDM2 absent (pleomorphic) means systemic disease. Infiltrative types (Ewing, DSRCT) need chemotherapy because no margin is achievable.

- † **FNCLCC grade** adds differentiation (1–3), mitoses per 10 high-power fields (1: 0–9, 2: 10–19, 3: ≥20) and necrosis (0 none, 1 <50%, 2 ≥50%): **2–3 = G1, 4–5 = G2, 6–8 = G3**. Grade and histology, not size, predict distant spread.

5 · Neurogenic tumours

Mostly benign, but MPNST and paraganglioma need special care.

Tumour	Key features	Management
Schwannoma	Paravertebral; encapsulated; S100+; slow	Local resection curative (often enucleable)
Neurofibroma	NF-1 or sporadic; malignant change up to 20%	Surgical
MPNST	<10% of RP sarcoma; half NF-1; H3K27me3 loss; spreads along roots	Resect with **parent nerve**; 5-year ~70% R0 vs ~20% R1
Ganglioneuroma	3rd–4th decade; calcification up to 30%	Excision
Neuroblastoma	MYCN adverse; MIBG staging	Paediatric chemotherapy with or without surgery
Paraganglioma	Commonest at organ of Zuckerkandl (below inferior mesenteric artery, IMA); T2-bright; secretory ~40%; metastases 20–50%; NF-1, MEN-2A, SDHB	Surgery after α-blockade† (see *Ch 145*)

6 · Germ cell, haematological, cystic, unknown primary

Rule these out before planning a sarcoma resection.

- **GCT**: an RP mass in a young man is GCT until proven otherwise. Extragonadal disease (3–5%) is often a **burnt-out testicular primary**. Give platinum first, then resect all nonseminomatous residual masses, because teratoma is chemoresistant and marker-silent. Marker: i(12p). See *Ch 79–80*.
- **Sex cord stromal** tumours: ~10% metastasise and they resist chemo and radiotherapy, so do early RPLND in high-risk node-negative primaries (age, >4–5 cm, necrosis, mitoses, atypia, lymphovascular invasion).
- **Lymphoma** forms confluent nodal masses: biopsy, never resect. **Plasmacytoma**: M-spike, CD138.

- **Cystic lymphangioma** (male children, crosses midline), mucinous cystadenoma and cystic mesothelioma (women) need **complete excision**, because aspiration or marsupialisation recur.
- **Unknown primary**: the primary is found in <20%. If midline or i(12p)-positive, give platinum; a single deposit is excised or irradiated.

> **KEY POINT**
> Histology decides first treatment: exclude chemotherapy-first entities (GCT, lymphoma, Ewing, midline unknown primary) before planning sarcoma resection.

7 · Surgery for RP sarcoma

The first operation decides cure.

- **Complete en bloc (R0) resection at the first operation** in an experienced centre; † it is the best chance.
- **Compartmental resection** removes adherent uninvolved neighbours (kidney, colon, psoas fascia), which lowers local recurrence. Preserve duodenum, pancreatic head, major vessels and femoral nerve unless invaded. † **Never enucleate**, because the pseudocapsule contains tumour.
- Nephrectomy is needed in up to **55%** but rarely leads to dialysis; † get a split renogram first. ~5% lose the femoral or obturator nerve and need a knee brace.
- CT within 6 weeks; no bowel prep. In GCT, check markers within 7–10 days and bank sperm. After **bleomycin**, keep FiO_2 low and fluids restrictive.
- Complete IVC thrombosis simplifies surgery (collaterals exist; resect); partial thrombosis may need a PTFE or allograft.
- Midline laparotomy; mesenteric root or Toldt mobilisation. **Split and roll** over the great vessels; ligate lumbar vessels and retrocrural lymphatics (to prevent chylous ascites); protect sympathetic trunks and the genitofemoral nerve.
- The **aorta is always reconstructed**. A chronically occluded **IVC** tolerates ligation; IVC grafts thrombose in 10–40%, so anticoagulate for ≥3 months.

8 · RPLND in germ cell tumours

Templates follow testicular landing zones.

- P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — *landing zones, templates, PC-RPLND, nerve-sparing*
- Primary RPLND gives survival ~100% with either template. In post-chemotherapy RPLND (PC-RPLND), a full bilateral infrahilar template is standard; unilateral only if the mass is **≤5 cm** in the landing zone, markers are normal, and IGCCCG risk is good or intermediate.
- Sparing the L1–L4 fibres and the superior hypogastric plexus below the IMA gives antegrade ejaculation in **90–100%**; temporary anejaculation is neuropraxia.

9 · Complications

Complications follow the dissection.

- P1 Lymphocele and postoperative lymphatic leak (chylous ascites) — *risk, diagnosis, stepwise treatment*

Not in the paper answer:

- Shunt or ligation when output stays >100 mL/day; MCT diet for 1–3 months after resolution.

- Pancreatic leak, VTE after caval reconstruction, nerve injuries.
- Chylous ascites occurs in 2–7% after chemotherapy (suprahilar dissection, IVC resection). Left suprahilar work can cause a pancreatic leak, so check drain amylase. Venous thromboembolism (VTE) follows ~1 in 5 caval reconstructions. Retractors cause femoral neuropraxia; combined mediastinal and RP resection can cause paraplegia.

> ALGORITHM
> **Chylous ascites**:
> 1. Paracentesis, a medium-chain triglyceride (MCT) diet, with or without octreotide.
> 2. If it reaccumulates, place a drain.
> 3. Then nil by mouth and total parenteral nutrition.
> 4. If output stays >100 mL/day, then peritoneovenous (LeVeen) shunt or ligation.
> 5. Continue the MCT diet for 1–3 months after resolution.

10 · Adjuvant and neoadjuvant therapy

Neither is routine; surgery is the mainstay.

- **Radiotherapy (RT)**: pre-op is better tolerated than post-op, where 50–60 Gy causes bowel toxicity. **STRASS** showed no gain, so it is **not standard**; † an exploratory benefit in (low-grade) liposarcoma keeps it an option. Intraoperative RT and brachytherapy are investigational.
- **Chemotherapy**: WD liposarcoma and leiomyosarcoma are largely resistant; myxoid liposarcoma, synovial sarcoma and Ewing respond. **Standard neoadjuvant chemotherapy is only for Ewing**. If resectability is doubtful, give induction multimodal therapy.

11 · Recurrence and follow-up

The pattern of failure differs by histology.

- † Local recurrence kills in liposarcoma; leiomyosarcoma fails distantly (lung, liver).
- † CT every 3–6 months for 2–3 years, 6-monthly to 5 years, then annually; lifelong for liposarcoma.
- Re-resect if resectable and slow (WD liposarcoma); each later R0 is less likely.

12 · Landmark trials

STRASS defines current practice on pre-op radiotherapy.

Trial	Design	Result	Take-home
EORTC-62092 STRASS	Phase III, 266 patients: pre-op RT + surgery vs surgery	No abdominal recurrence-free survival difference (HR 1.01); more complications	Pre-op RT not standard
STRASS II (ongoing)	Neoadjuvant doxorubicin + ifosfamide/dacarbazine + surgery vs surgery	Awaited	Neoadjuvant chemotherapy unproven

★ Putting it together: adrenal and retroperitoneal decisions

A · Adrenal incidentaloma: two questions

Skeleton of any adrenal-mass answer.

ALGORITHM

1. **Malignant?** If unenhanced CT is ≤10 HU or MRI shows drop-out, then adenoma. If >10 HU, do washout CT (APW >60% / RPW >40% = adenoma). If >4 cm, suspicious or growing >1 cm, consider resection (*Ch 146*).
2. **Functioning?** Test all: 1-mg dexamethasone suppression test (DST) and metanephrines; aldosterone-to-renin ratio (ARR) if hypertensive or hypokalaemic.
3. If **silent, <4 cm and benign**, then no follow-up.

B · Functional tumours compared

Screen, localise, prepare, operate.

	Aldosteronism	Cushing	Phaeo
Screen	ARR	1-mg DST, late-night salivary cortisol, urinary free cortisol	Metanephrines
Next	Confirm, then CT, then adrenal vein sampling	ACTH: low = adrenal; normal/high = pituitary/ectopic	CT/MRI, functional imaging; genetics for all
Pre-op	Correct K^+	Steroid cover	α then β; fluids, salt
Post-op	Stop K^+/spironolactone; hyperkalaemia	Steroids until axis recovers (months)	Hypotension, hypoglycaemia
Medical	Mineralocorticoid antagonist (bilateral)	Steroidogenesis inhibitors	Long-term α-blockade

C · Adrenalectomy approach

Suspected cancer is what changes the approach.

KEY POINT

Use laparoscopy (transperitoneal or retroperitoneoscopic) for benign tumours. Operate open for large suspected adrenocortical carcinoma or invasion, because rupture spills tumour and causes recurrence. On the right, the short vein enters the IVC, so control it carefully. Partial adrenalectomy suits bilateral or hereditary disease (*Ch 147*).

D · Retroperitoneal mass

Exclude non-surgical diagnoses first.

ALGORITHM

1. **Image**, then **exclude** paraganglioma (metanephrines), germ cell tumour (testis, markers) and lymphoma before biopsy.
2. If it changes treatment, do a **core biopsy** by the retroperitoneal route.
3. **Sarcoma:** en bloc R0 resection at the first operation (multivisceral if needed) in a specialist centre; never enucleate. Follow with lifelong CT, especially for liposarcoma (*Ch 140*).